

- 5 A 6-sided dice, A , with faces numbered 1, 2, 3, 4, 5, 6 is biased so that the probability of throwing a 6 is $\frac{1}{4}$. The random variable X is the number of 6s obtained when dice A is thrown twice.

(a) Find the probability generating function of X . [2]

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A second dice, B , with faces numbered 1, 2, 3, 4, 5, 6 is unbiased. The random variable Y is the number of 6s obtained when dice B is thrown twice.

The random variable Z is the total number of 6s obtained when both dice are thrown twice.

(b) Find the probability generating function of Z , expressing your answer as a polynomial. [3]

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(c) Find $\text{Var}(Z)$.

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(d) Use the probability generating function of Z to find the most probable value of Z .

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